AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M140
Introducing Statistics

Final Examination
Spring Semester 2025-2026

Date: 11/06/2026

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions:5
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 50% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

AOU EGYPT

Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 2 marks, giving a total of 10 possible marks.

Q–1: Which of the following can't be the probability of an event?

a) 0 **c)** 1.5
b) 1/4 **d)** 1

Q–2: Eye color is an example of what level of measurement?

a) ordinal **c)** ratio
b) nominal **d)** interval

Q–3: If a variable assumes the values 1, 4, 6, and 9, then this variable is a ..... variable.

a) continuous **c)** discrete
b) continuous and quantitative **d)** discrete and qualitative

Q–4: What is the statistical term for the mean that is obtained by using all data values of an entire population?

a) parameter **c)** variable
b) statistic **d)** quantity

Q–5: The standard normal distribution has a mean of ..... and a standard deviation of .....

a) 1 and 0 **c)** 1 and 1
b) 0 and 1 **d)** 0 and 0

Part 2: ESSAY QUESTIONS

Each question is worth 10 marks. You should answer all FOUR questions, giving a total of 40 possible marks.

Q–1: [5+5 Marks]

a) Consider the following sets of data:

A: 2, 4, 6, 8, 10 B: 2, 6, 6, 6, 10

(I) Find the standard deviation of each set of data
(II) Which of the two sets of data is more dispersed?

b) The following table gives 500 students classified by gender and grade.

If a student is selected at random, find the probabilities of the following events:
(i) P(Pass ∩ Male)
(ii) P(Male ∩ Female)
(iii) P(Male ∪ Female)

	Pass (P)	Fail (F)	Total
Male (M)	200	50	250
Female (F)	210	40	250
Total	410	90	500

FOR EXAM

Q-2: [10 Marks]

To check the existence of a relationship between exercise habits and having heart disease, a sample of 100 was chosen and the following table was obtained:

	Having Heart Disease	Normal
Rarely Exercise	14	6
Sometimes Exercise	11	10
Regularly Exercise	5	54

We want to use the sample data to decide whether there is an association between exercise habits and having heart disease at 0.05 level of significance.

[Given: $\chi^2_{0.05,2} = 5.991$]

Q–3: [5+5 Marks]

a) The studying times (in hours) of students preparing for an exam have a normal distribution with an unknown mean and standard deviation 4 hours. If the mean studying time of 16 students selected at random is 15 hours. Find a 90% confidence interval for μ.

[Given: $z_{0.05} = 1.645$]

b) The weights (in kg) of workers in a certain factory are normally distributed with mean 160 kg and standard deviation 8 kg. What proportion of workers have weights between 152 and 176 kg?

Hint: Use the table below

z	-1	-2.00	1.17	1.18	1.60	2.00	1.80	3.00
P (Z < z)	0.1587	0.0228	0.8790	0.8810	0.9452	0.9772	0.9641	0.9987

Q–4: [10 Marks]

A food company produces cans of juice that have a fill volume that is approximately normally distributed with a mean of 800 ml and a standard deviation of 40 ml. Test the hypothesis that μ = 800 ml against the alternative μ ≠ 800 ml if a random sample of 30 cans has an average fill volume of 788 ml at the 0.05 level of significance.

[**Given: $Z_{0.025} = 1.96$**]

End of Questions

ROUGH